Modeling Multi-Granularity Context Information Flow for Pavement Crack Detection

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***Abstract*— Pavement cracks have a highly complex spatial structure, a low contrasting background and a weak spatial continuity, posing a significant challenge to an effective crack detection method. To precisely localize crack from an image, it is critical to effectively extract and aggregate multi-granularity context, including the fine-grained local context around the cracks (in spatial-level) and the coarse-grained semantics (in semantic-level). In this paper, we apply the dilated convolution as the backbone feature extractor to model local context, then we build a context guidance module to leverage semantic context to guide local feature extraction at multiple stages. To handle label alignment between stages, we apply the Multiple Instance Learning (MIL) strategy to align the feature between two stages. In addition, to our best knowledge, we have released the largest, most complex and most challenging Bitumen Pavement Crack (BPC) dataset. The experimental results on the three crack datasets demonstrate that the proposed method performs well and outperforms the current state-of-the-art methods. On BPC, the proposed model achieved AP 88.32% with the 16.89 M parameters under the 45.36 GFlops runing speed. Datset and code are publicly available at: https://github.com/pangjunbiao/BPC-Crack-Dataset.**

***Index Terms*— Crack detection, context information, multi-scale, spatial structure.**

I. Introduction

PAVEMENT crack is one of the most potential yet important factors that endanger road safety, seriously affecting the life of the infrastructure. The accurate detection of cracks is increasingly becoming a vital method to extend the life of roads. Efficient crack detection method is a practical requirement especially for bitumen pavement.

The core issue of crack detection is to achieve the precise crack localization and its area. As shown in Fig. 1, both the unique structure and the special appearance the cracks pose significant challenges. Specially, bitumen pavement as a mix of bitumen and gravels naturally forms a crack-like road surface as folows:

1) **Low contrast**: Tiny and unclear cracks have a low contrast with the background. Besides, the high similarity between the cracks and the background makes many cracks too difficult to be accurately labeled.
2) **Weak continuity**: The crack has a group of structures with a irregular and weak continuity distribution. It indicates that cracks would occur at any position on roads.
3) **Noise interference**: The background contains complex noises, *e.g.*, the spots with different shapes and the scratches with varying lengths. These crack-like noises seriously affect the performance of crack detection methods.

Both semantic segmentation [1], [2] and object detection [3], [4] are the two mainstream methods for the crack detection task. However, due to the unique spatial structure of cracks, we find that both methods have their drawbacks. For the segmentation approach, pixel-level result guarantees the precise crack detection, but it requires the pixel-level annotations which cause a significant amount of time consumption. As shown in Fig. 1 (a), cracks in the red rectangle are almost impossible to be accurately labeled in a pixel-wise level. For the object detection method, it requires a bounding box to label the crack positions. However, the bounding box barely reflects the weak continuity of cracks, being unable to accurately measure the detailed cracks.

Compared to the semantic segmentation approach and the object detection one, as illustrated in Fig. 1 (b), patch-level annotation balances between efficiency and accuracy. It seems that if a patch is considered as a pixel, The approaches from the semantic segmentation community can readily be used for the crack detection. However, the patch-level annotation makes the approaches for the semantic segmentation inefficient for crack detection. For instance, the patch-level annotation barely generates the accurate ground truths for the different scales for U-Net [2].

In this paper, we propose to fuse **M**ulti **G**ranularity context information to model a **Crack** detection neural **Net**work (MGCrackNet), as illustrated in Fig. 2. MGCrackNet firstly leverages the dilated convolution to extract local context information for cracks, where local features (*e.g.* textural details of roads) and the spatial information is captured to precisely locate cracks.

Deep layers have more semantic information than that of the shallow ones [5]. To leverage this observation, a natural idea is that the shallow layers need to be guided by the semantic information from the deep layers. Concretely, we propose the frozen training strategy to force the multi-granularity context information from the deep layers to the shallow ones. Besides, we introduce Multiple Instance Learning (MIL) strategy [6], [7], [8] to align the context cues from the the deep layers to

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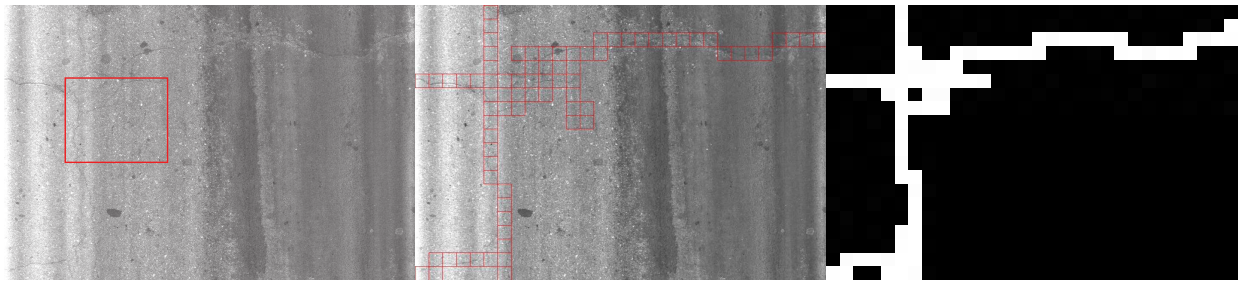

(a) A sample (b) Result of our method (c) Binarized ground truth

Fig. 1. A result of the proposed method on BPC. Cracks in the red rectangle are difficult to be labeled in Fig. 1(a), and please zoom out the image for more details. The red patches in (b) indicate cracks. (c) is a binary image of cracks. Best viewed in color.

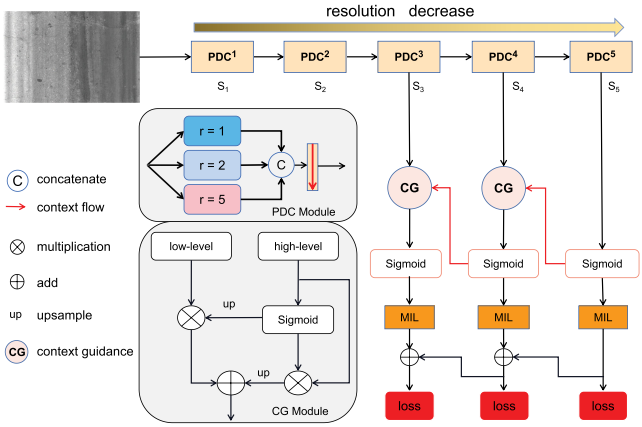

Fig. 2. The network structure of MGCrackNet, where 5 PDC modules extract the crack features, 2 CG module models the contextual information, and MIL module solves the LAP problems.

shallow ones. Comparative experiments demonstrate the the superior ability of our method. The contributions of this work are summarized in the two-fold:

- We firstly propose multi-granularity context information flow to gradually guide context cues from deep layers to shallow ones. Besides, a frozen training strategy is proposed to efficiently fuse context cues and local ones.
- To our best knowledge, we release the largest, the most complex and challenge bitumen pavement crack dataset. It brings new and considerable challenges for the crack detection community.

II. Related Work

A. Pixel-Level Crack Detection

Pixel-level crack detection is the semantic segmentation-based method that aims accurately to classify the crack pixels. The key to crack pixels classification lies in distinguishing between cracks and background. For example, the physical property was a straightforward way for crack detection including spatial continuity [9], [10], [11]. Leveraging context information was an efficiency method [12], [13], [14], [15]. For example, transformer [16], [17] were used to utilize the global context information. However, most of these cracks are serious cracks from the cement pavement which is easily labeled.

Due to the complex spatial structure of cracks, it is difficult to label cracks in a pixel-wise approach. To handle the difficulty of labeling, transfer learning [18], [19], [20] based approach were proposed to reduce the number of correctly labeled samples. In practice, the industrial requirement pays attention to these fine-grained cracks, because the repair cost for the tiny cracks is lower than that of the serious ones.

1) Region-Level Crack Detection: Region-level crack detection is the object detection-based method that essentially judges whether there is a crack or not. For example, You Only Look Once v3 (YOLOv3) was combined with deformable convolution to locate cracks [4], [21]. The method combined attention models into YOLOv5 for crack detection [3], [22]. In addition, [23] used fast R-CNN to detect cracks to handle the different weather conditions and the lighting levels. Reference [24] used Mask R-CNN coarse locate cracks to determine the severity of crack damage. The performance of the region-level crack detection is continuously improved with the progress of the object detection community. However, the predicted results are too imprecise to quantify the area of cracks.

2) Patch-Level Crack Detection: Patch level crack detection is a classification method that divides a crack image into multiple patches and further determines which patch contains a crack. In [25], a deep CNN was proposed to classify the image patches. In [26] and [27], context information from the different layers were used to improve the predicted results. In [28], the optimization of crack characteristics at each stage improved the crack detection performance, adopting the idea of modeling context information. However, the above methods rely on feature-level context and barely utilize the semantic contextual information. On the contrary, our Context Guidance (CG) module not only utilizes the context cues but also refines the feature extraction with the help of the MIL-based label alignment.

III. Modeling Context Information

A. MGCrackNet

In this paper, we adopt the parallel dilated convolution(PDC) [29] module to extract local context of cracks. As shown in the PDC module in Fig. 2, the PDC module is composed of 3 dilated convolutional layers with the dilated rates of $r \in \{1, 2, 5\}$ as follows:

$$O_r = C_r(F_i), \tag{1}$$

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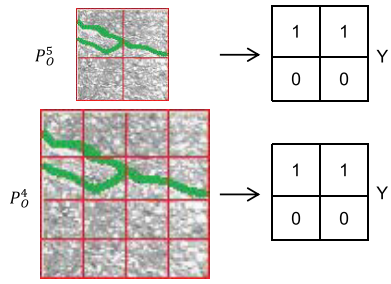

Fig. 3. The illustration of LAP. The green represents the crack, 1 indicates whether the patch contains a crack or not. We need to expand the 2×2 label Y into the 4×4 one for the stage S_4.

where F_i is the feature, and $C_r(\cdot)$ is the convolution operator with the dilated rate r. Dilated convolution in (1) is the same as convolution but it involves pixel skipping, so as to cover a larger area of receptive field without stacking too deep networks. However, these holes result in the loss of information. Therefore, we concatenate the paralleled features, *i.e.*, $S_o = concat(O_1, O_2, O_5)$, to obtain the different angularities contextual information for the pavement cracks.

As shown in Fig. 3, in a cracked patch, only a small number of pixels belong to cracks, while the other regions are background or noises. Consequently, we apply the max-pooling to sub-sample the result of PDC (1). We further group PDC and max-pooling as the module of the backbone in our network, as shown in Fig. 2.

MGCrackNet further separates the backbone of network into different stages.

Definition 1: (Stage). The Stage is a repeatable feature extraction module in a neural network. Usually, a stage consists of the combination of the convolution-pooling operations. In this paper, we split the network into 5 stages S_i, $(i \in \{1, 2, 3, 4, 5\})$. Features extracted from 5 stages are respectively donated as $F_{S_i} \in \mathbb{R}^{\frac{1}{2^i}H \times \frac{1}{2^i}W \times C_i}$, $(C_i \in \{32, 64, 128, 256, 256\})$, $(i \in \{1, 2, 3, 4, 5\})$, where H and W are the height and width of an input image, respectively, as well as C_i is the number of channels for the i-th stage.

The previous approach [28] stacks several backbones of the network into a new network to boost the performance. However, stacking multiple backbones naturally lead to an over-load yet inefficient neural network during the training and the deployment stages. On the contrary, our stage-wise approach is more efficient than that of the backbone-wise approach [28].

In this paper, we only use the feature maps of the last 3 stages, $F_{S_3} \in \mathbb{R}^{\frac{1}{8}H \times \frac{1}{8}W \times C_3}$, $F_{S_4} \in \mathbb{R}^{\frac{1}{16}H \times \frac{1}{16}W \times C_4}$ and $F_{S_5} \in \mathbb{R}^{\frac{1}{32}H \times \frac{1}{32}W \times C_5}$. Because the last 3 stages contain less noise interference [30]. $F_{S_{i-1}}$ represents the local context of cracks, but includes more noise interference [31]; while, F_{S_i} contains the high-level semantic information at the cost of losing spatial context of cracks. Therefore, a reasonable approach should be proposed to combine the advantages of both high-level features and the low-level ones.

B. Context Guidance Module

The CG module leverages the semantic context to help the shallower stage to extract the informative features and the local context, as illustrated in Fig. 2. Denote the features extracted by the stage S_i as F_{S_i}, the corresponding classification heatmap M_{S_i} is computed as follows:

$$M_{S_i} = \sigma(F_{S_i}), \tag{2}$$

where $\sigma(\cdot)$ is the sigmoid function, i indexes the stage $i \in \{3, 4, 5\}$, $M_{S_i} \in \mathbb{R}^{H_{S_i} \times W_{S_i} \times C_i}$, and H_{S_i} and W_{S_i} are the size of feature map in the i-th stage, *i.e.*, $H_{S_i} = \frac{1}{2^i}H$, $W_{S_i} = \frac{1}{2^i}W$.

The heatmap M_{S_i} in (2) means the probability of whether a patch contains a crack or not. Therefore, the heatmap M_{S_i} acts as an attention mechanise. Besides, the features F_{S_i} in (2) contain the context cues of the consecutive stage. Because the receptive field of the stage S_i is larger than the stage S_{i-1}. We consequently enhance the discriminative ability of feature F_{S_i} as follows:

$$\hat{F}_{S_i} = M_{S_i} \otimes F_{S_i}, \tag{3}$$

where $\otimes$ is the element-wise multiplication. The weighted feature $\hat{F}_{S_i}$ is a self-refined feature.

To align the feature size between two consecutive stages, the heatmap M_{S_i} is upsampled through bilinear interpolation operation. The feature $F_{S_{i-1}}$ is weighted by the updampled heatmap $up(M_{S_i})$ as follows:

$$\hat{F}_{S_{i-1}} = up(M_{S_i}) \otimes F_{S_{i-1}}, \tag{4}$$

where $up(\cdot)$ is the upsampling function.

If we consider the heatmap M_{S_i} as the semantic context of the stage S_{i-1}, $\hat{F}_{S_{i-1}}$ is the context guided features. That is, in (4), the element-wise multiplication guides feature $F_{S_{i-1}}$ to pay attention to the important positions in the receptive field of the shallow stage. To verify this assumption, the gradient of feature $\hat{F}_{S_{i-1}}$ with respect to parameters θ as follows:

$$\frac{\partial \hat{F}_{S_{i-1}}}{\partial \theta} = \frac{\partial up(M_{S_i})}{\partial \theta} \otimes F_{S_{i-1}} + up(M_{S_i}) \otimes \frac{\partial F_{S_{i-1}}}{\partial \theta}, \tag{5}$$

where $\frac{\partial F}{\partial \theta}$ is the gradient of the features F with respect to parameters θ.

Obviously, in (5), the gradient $\frac{\partial \hat{F}_{S_{i-1}}}{\partial \theta}$ is controlled by four terms, *e.g.* M_{S_i}, $\frac{\partial up(M_{S_i})}{\partial \theta}$, $F_{S_{i-1}}$, and $\frac{\partial F_{S_{i-1}}}{\partial \theta}$, in which the heatmap M_{S_i} is generated by the feature $F_{S_{i-1}}$ from the previous stage S_{i-1}. If we simultaneously optimize $F_{S_{i-1}}$ and M_{S_i}, a possible gradient contradicts would happen between $F_{S_{i-1}}$ and M_{S_i}. Consequently, the inaccurate heatmap M_{S_i} tends to fail guiding the features for the shallow stages. Therefore, a natural yet effective strategy is firstly to froze $F_{S_{i-1}}$ and optimize M_{S_i} as follows:

$$\frac{\partial \hat{F}_{S_{i-1}}}{\partial \theta} = \frac{\partial up(M_{S_i})}{\partial \theta} \otimes F_{S_{i-1}}. \tag{6}$$

To retain the sufficient priors from the deeper stage and leverage the features from the shallow one, the upsampled feature $up(\hat{F}_{S_i})$ is element-wisely added to the features $\hat{F}_{S_{i-1}}$ from the shallow stage as follows:

$$F_{S_i}^o = add(up(\hat{F}_{S_i}), \hat{F}_{S_{i-1}}), \tag{7}$$

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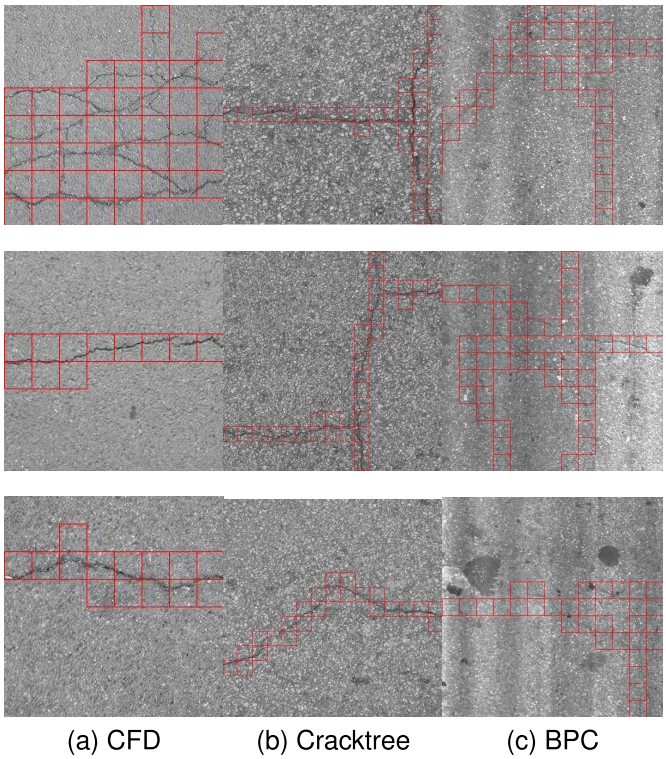

Fig. 4. Some samples from three datasets. The red patches coarsely mark the location of the cracks.

where $F^o_{S_i}$ is the weighted low-level feature at the stage S_i, and $add(\cdot,\cdot)$ is the element-wise addition operation.

C. Label Alignment for Stage-Wise Loss

If we supervise network at the different stages, we need to generate the corresponding annotation from the patch-level labels. However, the Label Alignment Problem (LAP) naturally happens since the ground truth is at the patch-wise granularity rather than the pixel-wise one. As shown in Fig. 3, the annotation of the sub-patches cannot generated from the 2×2 annotation for the stage S_4.

We apply the MIL strategy to solve the LAP. MIL is weakly supervised learning where training data is arranged in "bags" in which each bag contains a set of "instances" assuming that individual labels exist for the instances within a bag, but they are unknown during training. A bag is positive, if at least one instance in this bag is positive; otherwise; a bag is negative. Concretely, as illustrated in Fig. 3, a patch in the S_5 is a "bag", while the corresponding 2×2 sub-patches are "instance"s in the bag.

A Feature map $F^o_{S_i}$ is computed by a 3×3 standard convolution and two 1×1 standard convolutions for each sub-regions. The classification probability P^i_o for each sub-patch is computed as follows:

$$P^i_o = \sigma(F^o_{S_i}), \tag{8}$$

where $\sigma(\cdot)$ is the sigmoid function, and $P^i_o \in \mathbb{R}^{\frac{1}{2^i}H \times \frac{1}{2^i}W \times C_i}$ for the stage S_i.

Secondly, MIL strategy is used to convert the probability P^i_o of the sub-regions to the patch-wise annotation as follows:

$$Y^i_o = pool(P^i_o), \tag{9}$$

where $pool(\cdot)$ is the pooling function, Y^i_o is the classification probability at the different stages, $Y^i_o \in \mathbb{R}^{\frac{1}{32}H \times \frac{1}{32}W \times C}$. In MGCrackNet, the function $pool(\cdot)$ uses stride {4, 2, 1} for the S_i, respectively. During inference, if the probability Y^i_o is greater than 0.5, it indicates that a patch contains cracks. As illustrated in Fig. 2 when MIL is applied to the different stages, MGCrackNet effectively integrates the context information at varying levels of granularity.

D. Loss Function

We employ a multi-stage supervised approach to optimize the predicted results, as shown in Fig. 2. Concretely, given a labeled sample $\{X, Y\}$, where $X \in \mathbb{R}^{\frac{1}{2^i}H \times \frac{1}{2^i}W \times C}$ is an image, $Y \in \mathbb{R}^{\frac{1}{32}H \times \frac{1}{32}W \times C}$ is the corresponding multi-stage label, and $i \in \{3, 4, 5\}$ are index of the different stages, Cross Entropy (CE) loss of a sample $\{X, Y\}$ is used at the different stages as follows:

$$L = \sum_{i=3}^{5} CE(Y^i_o, Y), \tag{10}$$

where $CE(\cdot,\cdot)$ is the cross entropy loss.

IV. Experiments

A. Dataset Comparisons

1) CFD: The CFD [32] dataset is for the cement road crack. CFD contains 118 crack images with a size of 480×320. We use the 256×256 sliding window to crop the images, generating $2,832$ samples, where $2,256$ samples are used for the training and 576 ones are for the testing. Each sample is divided into 32×32 patches. Some samples from CFD are shown in Fig. 4 (a).

2) Cracktree: The Cracktree dataset is also for the cement road crack [33]. This dataset consists of 206 crack images with a size of 800×600. We use the 512×512 sliding window to generate the $1,300$ training samples and the 324 testing ones. Each samples is further divided into 32×32 patches. Some samples from Cracktree are shown in Fig. 4 (b).

3) BPC: We release the challengeable Bitumen Pavement Crack (BPC) dataset, which was collected from the professional road condition detection vehicle. The vehicle is certified by SCANNER organized by the Transport Research Institute in the UK. The vehicle is composed of a high-resolution linear array camera, with a scanning frequency of 36KHz/s, a detection width of 290 CM, and the continuous detection of small cracks with a size of about 1 mm. The images were captured from the highway from Beijing to Shijiazhuang in China, after snow-melting agent was used; therefore, the images have black-and-white stripes.

The dataset comprises $3,522$ images with a size of 640×960. Each image is divided into 16×16 patches with a size of 32×32. We use the 416×416 sliding window to generate $7,044$ samples, where the $5,764$ training samples and the

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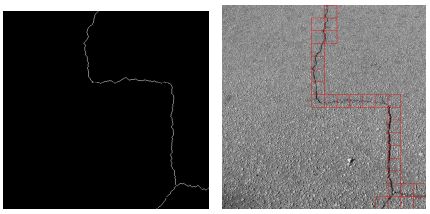
TABLE I
COMPARISON BETWEEN THREE DATASETS

Attributes	BPC	CFD	Cracktree
Material	Bitumen	Cement	Cement
Size	640×960	480×320	800×600
# Image	3522	118	206
# FGP	3080	0	0
$\frac{\#BGP}{\#CP}$	16 :1	3 :1	6 :1

FGP, BGP and CP denote Fine-Grained Patch, Back-Ground Patch, Crack Patch, respectively.



(a) pixel-level (b) patch-level

Fig. 5. An example of converting the pixel-level annotation into the the patch-level one.

1, 280 testing ones. Some samples from BPC are shown in Fig. 4 (c). BPC contains three types of cracks, *i.e.,* traverse, longitudinal, and alligator cracks. Note that no pre-processing is adopted to BPC.

Due to the complexity of bitumen pavement cracks, the cracks and background are highly similar. Therefore, BPC contains about 3, 080 "noisy" patches which are difficult to be correctly labeled by non-professionals (see Fig. 6). The noisy patches are caused by the materials (*e.g.* the mix of bitumen and gravels) of the bitumen pavement itself. We call them as the fine-grained cracks. During the labeling process, we have invited the experts to label these fine-grained patches.

We notice that there are several publicly accessible pavement crack datasets, *i.e.*, Crack500 [34] with 500 samples collected by cell phones, Aigle-RN [35] with 38 samples. However, these datasets supplied the limited number of samples. As summarized in Tab I, the number of samples, imbalance between the background and the foreground patches, as well as the difficulty of the fine-grained cracks, to our best knowledge, make BPC as the largest, most complex, and challenging bitumen pavement crack dataset to the SOTA methods.

It should note that both CFD and Cracktree are the pixel-level annotation, which does not meet our setting. During experiments, we regenerate patch-level labels from the pixel-level annotation as shown in Fig. 5.

B. Evaluation Metrics

In this paper, $Precision(P)$ and $Recall(R)$ are adopted to evaluate the performances among different methods. The $F1$ score measures the performance from P and R as follows:

$$F1 = 2\frac{P \cdot R}{P+R}, \tag{11}$$

where P is $Precision$, and R is $Recall$.

To further evaluate the performance of crack detection, $AveragePrecision(AP)$ of the cracks is computed as a comprehensive metric to reflect the overall stability of the model as follows:

$$AP = \int_0^1 P(t)dt, \tag{12}$$

where value $P(t)$ is the precision P at the threshold t, $t \in [0, 1]$. Therefore, AP is equivalent to the area under the precision-recall curve, measuring the performance of cracks rather than the background patches.

Algorithm 1 Freeze Training Strategy

INPUT: Features $\{F_{S_i}, i \in 3, 4, 5\}$
OUTPUT: Feature map Y_o^i, epoch index $t \leftarrow 1$
$Y_o^i \leftarrow F_{S_i}, i \in 3, 4, 5;$
for all $t = 1, \ldots, T$ **do**
  if $1 \leq t \leq 20$ **then**
    $Y_o^3, Y_o^4, Y_o^5 \leftarrow freeze(F_{S_3}), freeze(F_{S_4}), CG(F_{S_5});$
  else if $20 < t \leq 40$ **then**
    $Y_o^3, Y_o^4, Y_o^5 \leftarrow freeze(F_{S_3}), CG(F_{S_4}), CG(F_{S_5});$
  else
    $Y_o^3, Y_o^4, Y_o^5 \leftarrow CG(F_{S_3}), CG(F_{S_4}), CG(F_{S_5});$
  end if
end for
return $\{Y_o^i\};$

C. Experiment Settings

In our experiments, we use the open-source deep learning framework PyTorch 1.8.0, cuda 11.1, and run all experiments on an Intel(R) i7-7700 CPU @ 3.60GHz with 32 GB RAM and NVIDIA 3080. During training, we do not use any pre-trained models instead train the models from the scratch. We set the batch size as 16 and use Stochastic Gradient Descent (SGD). Weight decay and momentum are set to 0.0005 and 0.9, respectively. We set the initial learning rate to $1e{-}3$. After 40 epochs, it will be divided by 10 for every 20 epochs. we use data augmentation techniques (*e.g.*, random rotation, random cropping). to improve the model robustness. In addition, we employ the freezing training strategy in Alg. 1. We used the early stopping techniques to handle the possible overfitting problem.

D. Baselines and The State-of-Art Approaches

To demonstrate the superiority of the proposed method, we compare MGCrackNet with the SOAT methods on these public datasets from three aspects as follows:

- **Global Semantic Context**: Two general vision transformer networks, Swin Transformer [36] (ST) and Focal Transformer [37] (FT), are used as baseline to extract the global context features. Based on the Transformer [42] architecture, ST and FT extract global context features from samples and achieve the best performance in visual classification tasks. Therefore, we use both methods as baselines to extract global semantic context features for crack detection.
- **Local Context**: Three efficient architectures, VGG16 [40], ResNet50 [39] and DenseNet121 [38], are

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TABLE II
CONTRASTING EXPERIMENTAL RESULTS ON THREE DATASETS

Methods	BPC				CFD				Cracktree			
	$P(\%)$	$R(\%)$	$F1(\%)$	$AP(\%)$	$P(\%)$	$R(\%)$	$F1(\%)$	$AP(\%)$	$P(\%)$	$R(\%)$	$F1(\%)$	$AP(\%)$
ST [36]	0.00	0.00	0.00	9.16	86.90	72.56	79.08	86.98	85.66	85.58	85.61	90.09
FT [37]	0.00	0.00	0.00	9.25	80.30	67.01	73.05	81.92	78.03	75.42	76.70	84.23
DenseNet121 [38]	53.86	64.64	58.75	61.46	91.20	79.60	85.00	90.05	89.26	87.22	88.22	93.67
ResNet50 [39]	76.70	78.69	77.68	79.41	94.68	84.53	89.31	93.64	93.34	83.97	89.55	93.23
VGG16 [40]	77.95	**81.40**	**79.63**	82.51	95.57	83.77	89.28	94.76	**94.91**	84.77	89.55	93.23
DCNN [25]	74.46	68.44	71.32	75.84	94.32	85.13	89.48	94.83	91.51	79.01	84.80	91.94
MSCNN [26]	77.58	72.85	75.14	80.20	96.93	85.52	90.91	96.57	92.93	83.97	88.22	93.85
UFE [41]	76.23	80.17	78.15	82.37	96.96	90.09	93.39	95.76	92.44	82.76	87.33	92.63
SFE [28]	84.71	73.35	78.62	85.10	96.17	85.82	90.70	95.62	92.55	89.93	**91.22**	92.99
MGCrackNet	**86.90**	72.45	79.02	**88.32**	**97.21**	**90.35**	**93.65**	**97.29**	90.21	**90.17**	90.19	**95.24**

used as baseline to extract local context features. These algorithms are also successful in visual classification tasks.

- **The State of Art Crack Detection Methods**: The State-Of-The-Art (SOTA) methods includes Deep Convolution Neural Network (DCNN) [25], Multi-Scale classification network (MSCNN) [26], Structural Feature Extraction model(SFE) [28], and Unconventional Feature Extraction model (UFE) [41]. These methods belong to the patch-level crack detection algorithms. Except for DCNN, UFE, SEF and MSCNN utilize the local context to crack detection. The reasons that we list them as SOTA methods are as follows:
  - **DCNN:** This work detects cracks by a deep convolution network. However, this method ignores the importance of context information.
  - **UFE and SFE:** Both UFE and SFE extract local context information by feature fusing strategy. However, these methods did not handle label alignment problem.
  - **MSCNN:** This work uses multiple max-pooling operations at the different layers of the network to sub-sample the local context. However, these max-pooling operations cause a serious information loss.

  For a fair comparison, the learning rate and the other hyper-parameters of these STOA methods are tuned to obtain the best performances.

E. Experimental Results

1) Results on CFD: Tab. II shows that our method outperforms the other STOA methods in terms of both AP and $F1$. For instance, compared with DCNN, MSCNN, UFE and SFE, our method achieves 97.29%AP, which has 2.46%, 0.72%, 1.53% and 1.67% gains over STOA methods, respectively. We notice that the CFD barely has label noise and the cracks are almost clear ones for the cement pavement. As expected, the AP of MSCNN, UFE and SFE also achieve excellent performances in Tab. II. As a comparison, our method achieves competitive performances over two evaluation criteria from both $F1$ and AP metrics.

Another important observation is that UFE achieves the closest performance to our method in terms of $F1$, as illustrated in Fig. 7(b). Our method adopts the multi-stage context information flow to boost the discriminative ability of features. It indicates that multi-stage context cues is necessary for crack detection.

Either the global semantic context method (*i.e.*, ST and FT) or the local one (*i.e.*, DenseNet121, ResNet50 and VGG16) does not obtain the comparative performances in Table II. One possible reason is that the transformer based methods perform well for a large data sets in which the diversified local appearances help robustly use attention mechanism. However, the high similarity between the background and cracks damage this assumption, especially for the BPC dataset (*i.e.*, ST and FT obtains super inferior 9.16 and 9.25 AP, respectively).

2) Results on CrackTree: The Cracktree dataset is more complex than the CFD dataset, due to the shadows, and the blurred background.

Tab. II shows our method consistently achieves the best performance with an 95.24%AP. In addition, the transformer also achieve over 80% AP on Cracktree dataset, indicating that it can handle crack datasets with less noise. However, the transformer is unable to fully fit the data features due to insufficient datasets, resulting in a significant decrease in classification results compared to CNN. As illustrated in Fig. 7 (c), our method consistently outperforms both local features based methods and global feature based ones. It means the proposed context flow have advantage to fuse both local and global features.

3) Results on BPC: Tab. II shows the comparison results in terms of $F1$ and AP on BPC. There are two observations in Tab. II as follows:

- Our method achieves the best performances, *e.g.*, 88.32 % significantly outperforming the other methods. Specially, the AP of DCNN, MSCNN, SFE and UFE are 75.84%, 80.20%, 82.37%, and 85.10% respectively, which are lower than that of our method. Besides, as illustrated in Fig. 7 that, MGCrackNet holds a curve most close to the up-right corner, achieving the best precision and recall values.
- The recall of our method (*i.e.*, 72.45%) is a lower than that of UFE (*i.e.*, 80.17%), while the precision of our method (*i.e.*, 86.90%) is a higher than that of UFE (*i.e.*, 76.23%). One possible reason is that MGCrackNet could capture more useful context information than UFE. Besides, BPC has a lot of noisy and blurry cracks, and even some crack patches are difficult for humans to be

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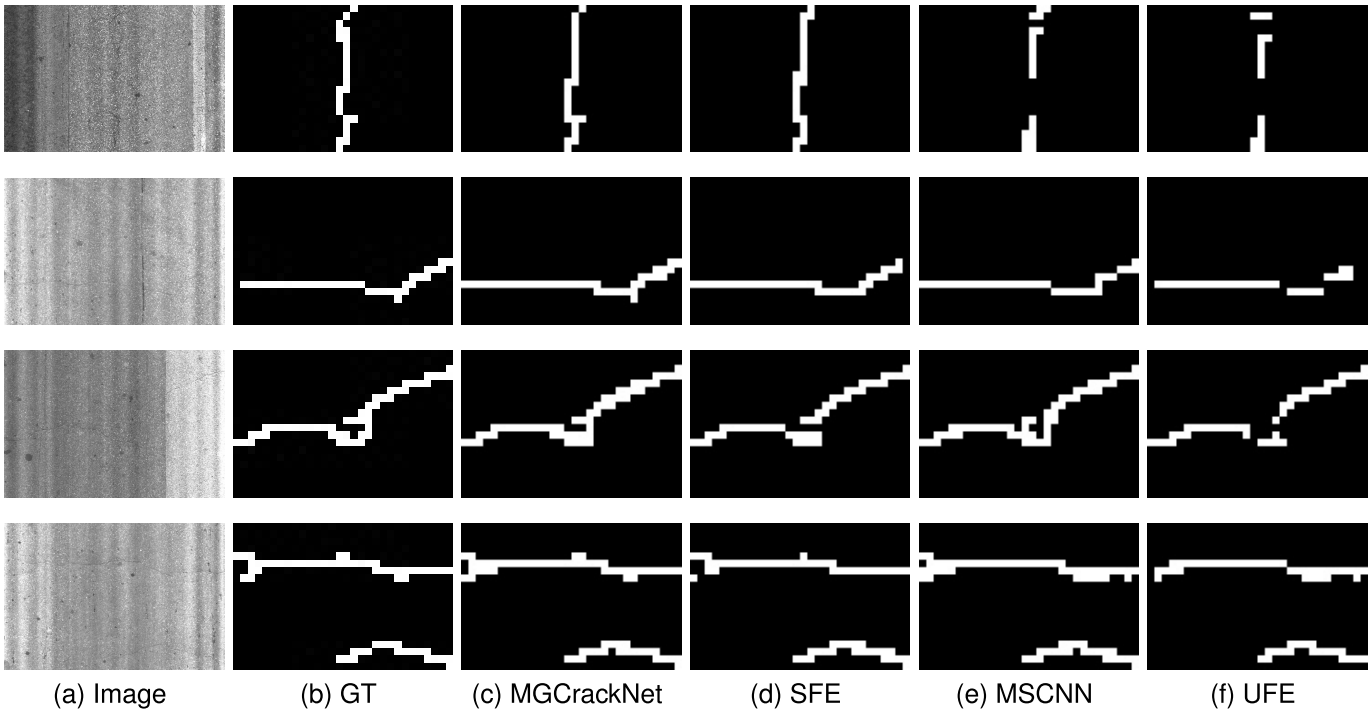

Fig. 6. Visual comparisons of the STOA methods and our methods on BPC. Please zoom out for more details. The white patches indicate the location of cracks, while the black block indicates the background.

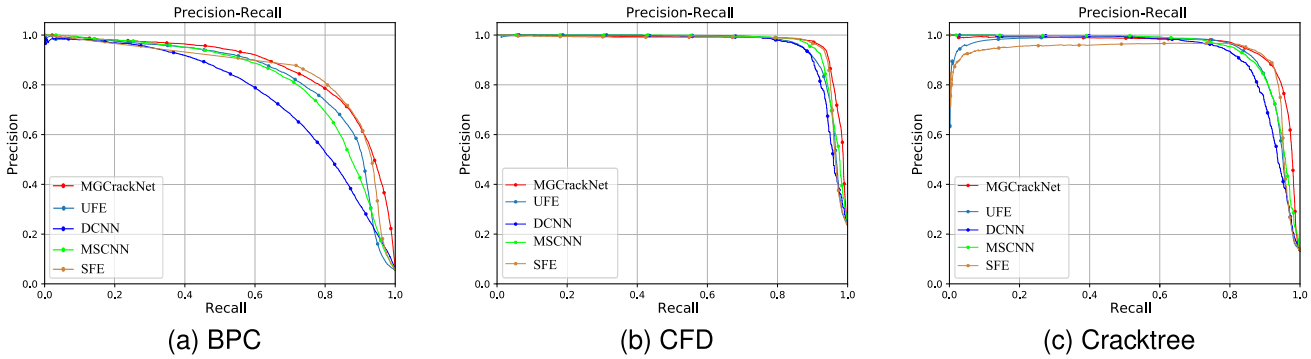

Fig. 7. The Precision-Recall curves of the crack detection methods on three datasets.

accurately labeled. Therefore, our method has a lower recall but higher precision to guarantee a higher AP than that of SFE, UFE, DCNN and MSCNN. Another possible reason is that although low-level features bring more detailed information, they also bring noisy information, especially for the results from the stage S_3.

F. Ablation Experiments

An interesting observation in Tab. II is that the AP of the three local context-based networks (*i.e.*, DenseNet121 [38] with 121 layers, ResNet50 [39] with 50 layers and VGG16 [40] with 16 layers) decline the performances with the increase of the network depth. It indicates that cracks do not requires a larger receptive field to extract the local context. Besides, the experimental results of transformer further validate the above explanation. The similarity between cracks and backgrounds make transformer [36], [37] inefficient for the BPC dataset. It indicates that boosting the discriminative ability of local features is more important.

TABLE III
ABLATION EXPERIMENTS ON BPC DATASET

SPDC	CG	MIL	Freeze	$P(\%)$	$R(\%)$	$F1(\%)$	$AP(\%)$
✓	-	-	-	76.23	**80.17**	78.15	82.37
✓	✓	-	-	85.22	71.03	77.31	86.62
✓	✓	✓	-	85.56	71.01	78.16	86.99
✓	✓	✓	✓	**86.90**	72.45	**79.02**	**88.32**

1) Ablation Experiments for Each Module: As shown in Tab. III, the combination of each module improves the results. Specifically, we add the CG module to the PDC based network, and the results show that the gain of AP increased by 4.25%, demonstrating the effectiveness of that the semantic context should guide the local context. In addition, the MIL module

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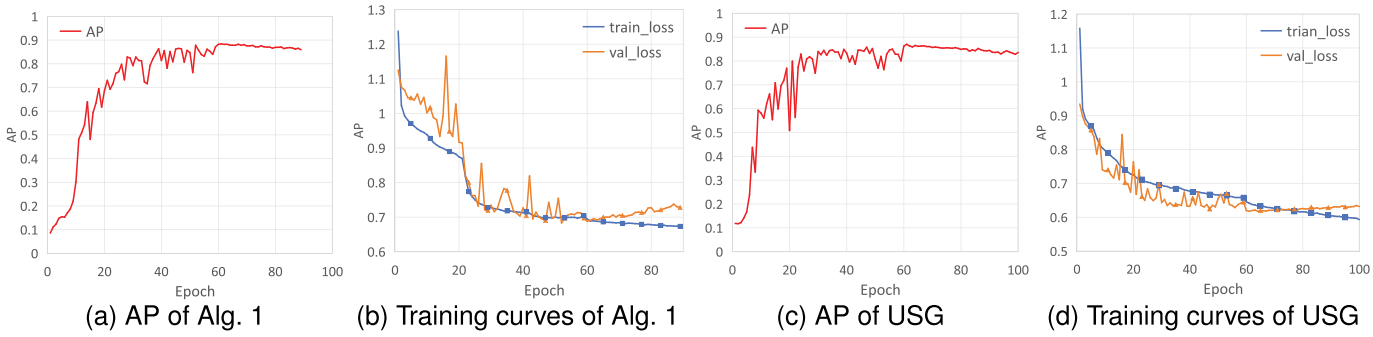

(a) AP of Alg. 1 (b) Training curves of Alg. 1 (c) AP of USG (d) Training curves of USG

Fig. 8. The comparisons between the frozen training strategy and the unfrozen one.

TABLE IV
THE PERFORMANCE OF THE DIFFERENT DILATED RATES ON BPC DATASET

Dilated Rates	$P(\%)$	$R(\%)$	$F1(\%)$	$AP(\%)$
{1,2,3}	86.27	71.22	78.02	86.75
{1,2,5}	**86.90**	72.45	**79.02**	**88.32**
{1,3,5}	85.06	72.22	78.11	86.96
{2,3,5}	80.46	**76.86**	78.61	86.37

TABLE V
THE PERFORMANCE OF THE DIFFERENT POOLING OPERATION OF MIL MODULE ON BPC DATASET

Pooling	$P(\%)$	$R(\%)$	$F1(\%)$	$AP(\%)$
Max	86.27	72.06	78.53	87.67
Avg	**86.90**	**72.45**	**79.02**	**88.32**

gains a 1.7% increment in terms of AP, because it further optimizes the noise interference of the output results.

2) Robustness of Hyper-Parameters in PDC: To select the most appropriate hyper-parameters of the dilated rate, we conduct a series of hyper-parameters experiments on the BPC dataset,as shown in Tab. IV. The results show that the hyper-parameter $r = 1, 2, 5$ has obtained the best experimental results.

3) Ablation Experiments for MIL Module: Tab. V experimentally selects the two most commonly used pooling methods: max-pooling(Max) and Average-pooling (avg). The avg-pooling achieves the higher experimental results than that of max-pooling, gaining 0.65%. The reason is that the max-pooling outputs the value with the highest probability, which only captures the information of one sub-patch. On the contrary, the avg-pooling back-propagates gradients for all sub-patches. That is, the continuous structure of cracks is a crucial factor to determine whether the patch contains cracks or not. Consequently, avg-pooling is more suitable for crack detection.

4) Ablation Experiments for Alg. 1: Fig. 8 compares between the frozen strategy in Alg. 1 and the Unfrozen StrateGy (USG) (*i.e.*, only line 10 in Alg. 1 trains MGCrackNet). Compared with Fig. 8 (d), Fig. 8 (b) decreases the training loss when the unfrozen steps were used (*i.e.*, epochs is at 20 and 40). Interestingly, although the USG in Fig. 8 (d) obtains more lower training loss and validation loss than these of the Alg. 1, the AP of frozen strategy in Fig. 8 (a) surpasses USG by 1.33% AP when the epoch is 60.

5) Running Time and Memory Cost: MGCrackNet has 16.89 million parameters. The training time and the memory footprint of MGCrackNet respectively are 71 seconds per epoch and 9.7 G, during the training stage. MGCrackNet costs 9.9 seconds to infer 1280 640 $\times$ 940 images.

6) Visualization: We visualize some predicted results by our model and other approaches in Fig. 6. The images from BPC dataset contain some noises and a complex crack structure. Crack map outputted by MGCrackNet is very close to the ground truth, showing an excellent performance. As a contrary, other methods (*i.e.*, SFE, UFE, and MSCNN) have discontinuous predicted results. In fact, the connection is very important to calibrate the damage degree of roads during the post-processing stage. These miss-classified patches are the tiny and unclear cracks which significantly pose the challenge to the crack detection community.

V. CONCLUSION

In this paper, we propose a MGCrackNet to handle the complex, irregular and non-discriminative cracks for the bitumen pavement roads. From the perspective of leveraging context cues, multi-granularity contextual information is progressively fused through the multi-stage CG process for crack detection. In addition, we apply the MIL strategy to handle label alignment problem. Experimental results on the three crack datasets show that the proposed MGCrackNet achieves an excellent performance, demonstrating the effectiveness of our method.

Modeling context information is still an essential approach to handle crack problems. We should introduce methods to deal with both the hard patches and the noisy labels. In the feature, how to handle noises and utilize unlabeled data are interesting directions. Besides, how to train an universal model that can detect various types of pavement cracks is an interesting problem from the viewpoints of self supervised learning and the large multi-modality model [43].

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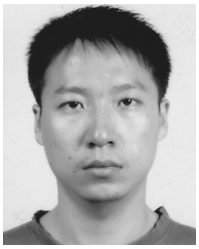

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